Task 1: QA & Debug Report

Conduit (RealWorld) demo app - exploratory testing of the main user flows

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App tested	conduit-realworld-example-app.fly.dev (public live demo)	**Browser**	Chrome + DevTools (Console, Network)

1. Summary

I tested the sign-up, login, logout and home-feed flows of the Conduit demo app and found **5 issues**: 1 Critical, 1 High, 1 Medium and 2 Low. The two that matter most for production are #3 (an account created with a short password can never log in) and #5 (the app shows a blank white screen when the server answers with HTTP 429). One issue, #3, is analysed in detail in section 4.

How I tested: manual black-box testing of each form with normal and edge-case input (1-character password, invalid email, duplicate email), plus browser DevTools to watch the network requests and console errors. Where a cause below is marked *suspected*, I inferred it from the observed behaviour and have not confirmed it in the source code.

2. Bug Table

#	Title	Steps to reproduce	Expected vs Actual	Severity	Suspected cause
1	**Junk and offensive tags shown in Popular Tags sidebar**	1. Open the Home page. 2. Look at the **Popular Tags** list in the right sidebar.	**Expected:** meaningful, clean tags. **Actual:** test/junk tags such as `fff`, `asd`, `tset`, `123` and an offensive word are shown to every visitor (Fig. 1).	**Low**	No validation or moderation of tag text when an article is created, so any string becomes a public tag.
2	**Weak password accepted at sign-up**	1. Open **Sign up**. 2. Enter a username and a valid email. 3. Enter password `1`. 4. Click Sign up.	**Expected:** error asking for a longer password. **Actual:** account is created with a 1-character password.	**Medium**	No minimum-length or strength rule on the sign-up form or on the server.
3	**Account created with a 1-character password cannot log in**	1. Sign up with password `1`. 2. Log out. 3. Open **Login**, enter the same email and password `1`. 4. Click Sign in.	**Expected:** either sign-up rejects the password, or login works. **Actual:** login form is blocked with the browser message "Please lengthen this text to 5 characters or more" (Fig. 2). The user is locked out of their own account.	**High**	The login password field has a 5-character minimum, the sign-up field has none, and the server accepted the short password. See section 4.
4	**Sign-up page shows "Sign in to your account"**	1. Open the **Sign up** page. 2. Read the text under the heading.	**Expected:** a link like "Have an account?" leading to Login. **Actual:** heading says "Sign up" but the link under it says "Sign in to your account" (Fig. 3).	**Low**	Wrong hard-coded text on the sign-up page (looks like copy-paste).
5	**Blank white screen when the server returns HTTP 429**	1. Open **Sign up**. 2. Submit the form repeatedly until the server replies with 429 Too Many Requests. 3. Look at the page and the DevTools Console.	**Expected:** a readable error message and the app stays usable. **Actual:** the whole page turns blank white. Console shows an AxiosError with status 429 on `/api/users`, followed by `Uncaught TypeError: Cannot read properties of undefined (reading 'headers')` (Fig. 4).	**Critical**	*Suspected:* the error-handling code in the app reads `.headers` on a value that is undefined in the 429 case (the request itself did fail with a normal 429 response), and there is no React error boundary to catch it. Exact line not confirmed (minified bundle).

Additional observation (not counted above, Low): the sign-up error "Email already exists.. try logging in" is clear, but it also tells anyone whether a given email is registered, which allows checking which people use the site (Fig. 3).

3. Screenshots

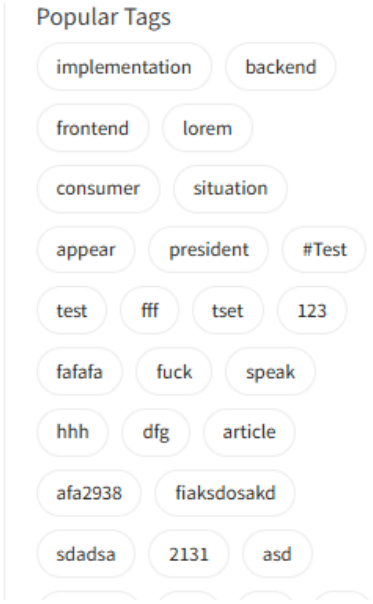


Fig. 1 - Bug #1: junk tags in Popular Tags.

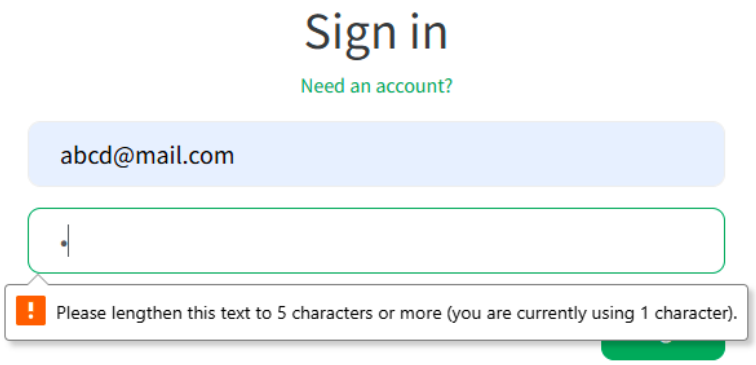


Fig. 2 - Bug #3: login blocked for the 1-character password.

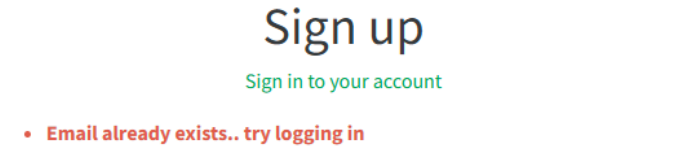


Fig. 3 - Bug #4 (wrong text under the Sign up heading) and the duplicate-email message.

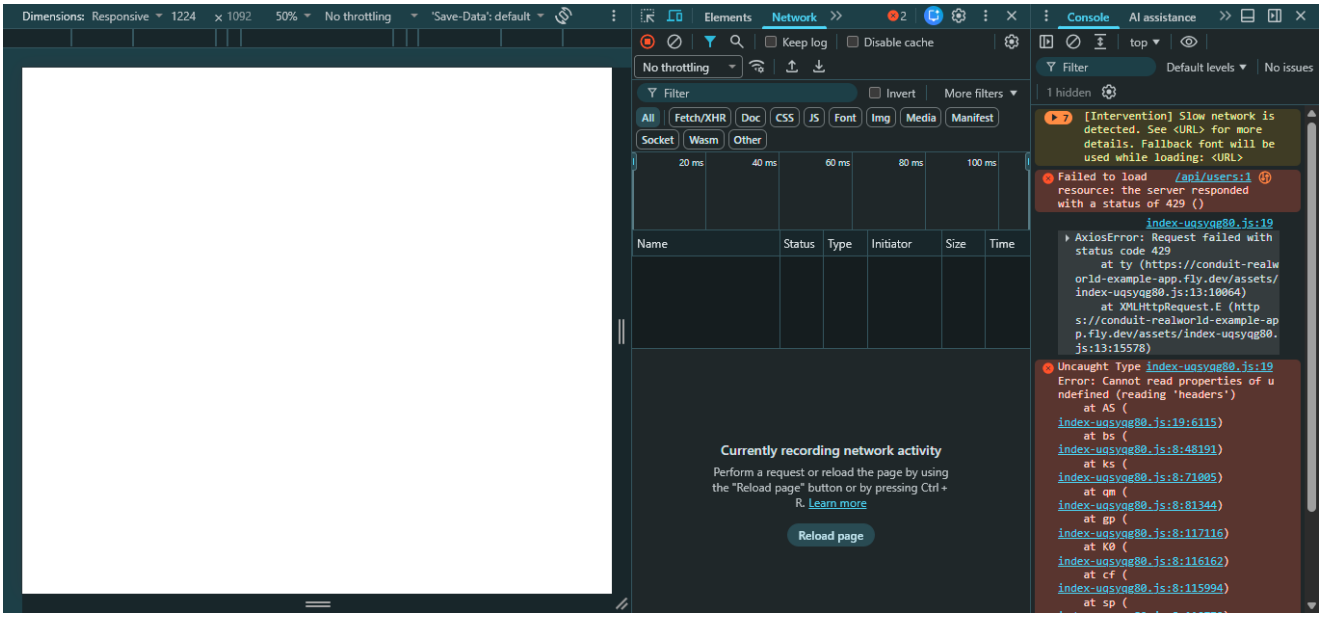


Fig. 4 - Bug #5: blank white page (left) and console errors after the 429 response (right).

4. Root-Cause Analysis - Bug #3 (account cannot log in)

What is happening. A user can sign up with the 1-character password `1`, but can never log in with it. Sign-up succeeds and creates the account. On the login page, however, the browser blocks the form with "Please lengthen this text to 5 characters or more" before any request is sent to the server.

Why. The two forms follow different rules. The message on the login page is the browser's built-in form validation, so the login password field must carry a 5-character minimum. The sign-up field has no such minimum, and because the server created the account, the server does not check password length either. The rule therefore exists in only one place: on the client, for only one of the two forms. (I inferred this from the behaviour; I have not yet confirmed it in the source code.)

How I would fix it. First, define one password policy (for example at least 8 characters) and enforce it on the server in the registration endpoint, returning a clear validation error. Second, apply the same rule on the sign-up form so the user gets instant feedback. Third, remove the length restriction from the login field: login should only check credentials and show "invalid email or password", never reject by length. Fourth, add tests for registering with a 1-character password (must fail) and for logging in with a valid account. Accounts that already exist with short passwords need a password reset.

```
// example server-side rule (express-validator)
body('user.password')
  .isLength({ min: 8 })
  .withMessage('Password must be at least 8 characters');
```